MACHINE LEARNING / APRENDIZAGEM COMPUTACIONAL – 2025/2026

PROJ2: CLASSIFICATION

Summary

- Apply and test classification techniques for static gesture recognition and facial recognition



The Project should be done in groups of 2/3 students. It will consist of 2 parts (read until the end).

Part I. – 75%
Using Media Pipe Hands (see refs [1],[2]) that returns hand landmarks, we propose to students to classify 6 **static gestures** of the right hand.

1. (10%) **To be done jointly with all students together.** For that, students will have to record their own data with a webcam. You should define a protocol to record the labelled data. The following are suggested (you can refine the process):
   - The video/images should record the upper body (head, right hand and trunk). Should work to Part-I and Part-II.
   - The gestures should be six: open palm hand, closed hand, V-shape hand, hand-to-right, hand-to-left, no-hand (or no gesture);
   - For each gesture, each student should record 6 clips of 10 seconds. To increase variability, the hand and the head should be in slightly different positions and distance (small controlled variations). The recording could be adjusted to 15 fps (set according to computer capacity);
   - The recorded videos (datasets) should be sequentially labelled/coded and organized in folders.
   - Split the data into training and testing, for example, 4 clips for training and 2 clips for testing, for each collected gesture (suggested: subject_id / gesture / clip_id).

2. (40%) **To be done in individual groups.** Start by extracting the hand landmarks and analysing the data. I suggest using only (x,y) coordinates, but you can also test (x,y,z).
   a) Apply classic Machine Learning Methods that were studied during the classes or others: SVM, KNN, and one additional method of your choice. Train and test offline;
   b) Apply a CNN-1D. Train and test offline;
   c) Test online the best solution for classical approach and for CNN-1D. I suggest to return a classification output every second or half second (by aggregating predictions over a temporal window, e.g., majority vote or mean probability);

3. (25%) **To be done in individual groups.**
   a) Apply a CNN-1D using a time-series window (e.g., 1 second). Train and test offline;
   b) Test online;

[1] - https://mediapipe-studio.webapps.google.com/studio/demo/gesture_recognizer
[2] - https://www.youtube.com/watch?v=pG4sUNDOZFg

Part II. – 25%
Using the recorded data obtained in part I, implement a facial recognition system. The system should be able to recognize the Students and the Professor.

1. Applying sequentially: "face detection + crop" + embeddings (e.g. FaceNet) + classic classifier. Train and test offline a recognition system. I suggest to start the implementation and testing on a public dataset.
2. Implement and test online;

Submission:

- One Jupyter notebook per part, with clearly identified sections/subsections. The Jupyter notebooks should indicate all dependencies and include a **requirements.txt / environment.yml** to build an environment. Confusion matrices and macro Precision are required for Part I. To submit on e-learning platform.
- A mini-paper regarding Parts I.2-3 and II.

Demonstration/presentation

- The best classical model and the best DL model for Part-I**,** and the facial recognition system for Part-II will have to be demonstrated and explained (to be scheduled).

Evaluation criteria

Conceptual and formal aspects (correct application of the methods, performance validation, and code optimization)	**40%**
Performance of the approaches and relative comparison between teams	**20%**
Jupyter notebooks presentation	**15%**
Mini-paper explaining methodology and results	**15%**
Global appreciation and demonstration/presentation	**10%**

Submission deadline Part I and Part II: 31st January 2026 (adjusted if required).
Part I.1 should be concluded until 16th January.

January 2026, Gabriel Pires